

Drawing the Cognitive Automation Boundary in Human-AI Teaming

A position paper on what AI should automate in knowledge work

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Abstract

Generative AI makes function allocation feel deceptively simple: if a model can perform an activity faster, the activity appears ready for automation. We argue that capability is the wrong primary criterion for knowledge work. Some activities are epistemically critical because personally reasoning through them builds or exercises the mental models, judgment, and transferable expertise a human will need later; other activities mainly instantiate an already-formed idea. We introduce epistemic criticality, $E(a|H)$, as the value of personally performing activity a for a particular human knowledge state H , and define the Cognitive Automation Boundary as the dynamic line separating cognition that should remain human-owned from execution that can be delegated. We distinguish an epistemic core, an epistemic-support layer, and an execution shell; show why the boundary must move with expertise; and propose a small decision framework combining epistemic criticality, execution burden, and automation reliability. Our position is that Human-AI teaming should maximize useful automation subject to preserving the cognitive operations required for future independent competence, supervision, and learning.

Keywords: Human-AI teaming; automation; knowledge work; epistemic criticality; cognitive offloading; learning; expertise.

1 Capability Is Not the Criterion

Human-AI teaming is often evaluated at the level of the coupled system: did the pair finish faster, produce a better answer, or reduce workload? Classic automation research likewise asks which functions should be automated and at what level [1]. These are necessary questions, but generative AI exposes a second allocation problem. When AI can write, analyze, design, code, visualize, and explain, the issue is no longer only which agent can execute a function. It is which cognitive operations the human must still perform in order to remain knowledgeable.

Consider building a knowledge-graph application. Deciding what counts as an entity, which relations carry meaning, what granularity is appropriate, and how an unexpected cluster should be interpreted are not merely production steps. They shape the builder's conceptual model of the domain. By contrast, converting CSV to JSON, generating repetitive CRUD endpoints, or moving a front-end button twenty pixels may be necessary to ship the artifact while contributing little to the knowledge the person actually needs to retain. Automating both classes simply because both are technically automatable erases this distinction.

Our position is therefore: a good division of labor minimizes unnecessary execution while preserving the cognitive operations necessary for human knowledge formation, judgment, and future independent competence. This extends calls to combine human agency with automation [2] by asking what kind of agency must remain cognitively exercised, not merely formally available.

Interactive concept: draw the Cognitive Automation Boundary

Drag the boundary in a future interface; the correct location depends on the human, task, and context.

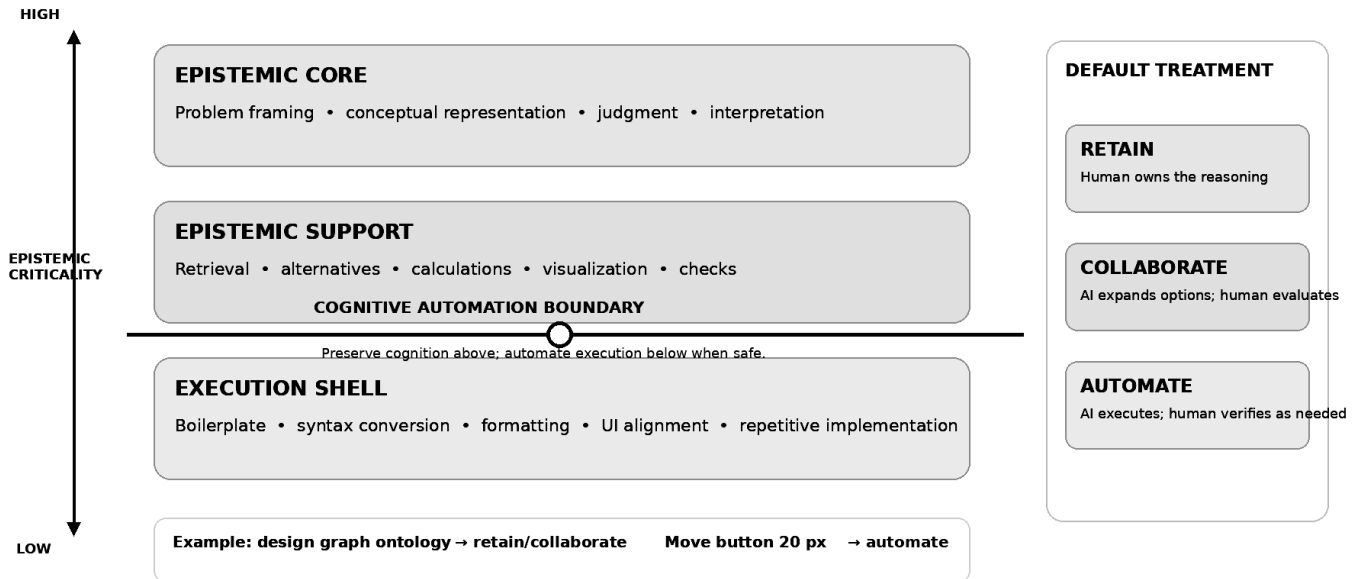


Figure 1. Interactive-style concept mock-up of the Cognitive Automation Boundary. Epistemically critical work stays above the boundary; low-criticality execution can move below it. The boundary is intentionally movable rather than universal.

2 Epistemic Criticality: What Work Builds the Human?

We define epistemic criticality $E(a|H)$ as the degree to which personally performing or reasoning through activity a contributes to the knowledge, mental models, judgment, or transferable expertise required by human H for future decisions. E is relational, not intrinsic. Writing a first SQL JOIN may be highly critical for a novice forming a relational model and nearly irrelevant for an expert who has written thousands. Debugging can be low-level implementation in one context and the main route to causal understanding in another. The unit of analysis is therefore activity x person x goal, not activity alone.

This framing is consistent with evidence that cognitive offloading changes the information-processing demands of a task [3], that excessive automation can leave people out of the loop with weaker situation awareness [4], and that self-generation can strengthen later memory compared with passive receipt [5]. None of these results implies that offloading is generally harmful. Rather, they show why the value of an operation cannot be inferred from its difficulty or tediousness alone.

3 A Three-Layer View of Knowledge Work

For design, we propose a layered heuristic. The epistemic core contains operations through which the human frames and represents the problem: defining concepts, selecting evidence, choosing abstractions, interpreting anomalies, and making consequential judgments. AI may challenge, compare, or propose alternatives here, but should not silently replace the human's reasoning. Epistemic support contains retrieval, calculation, simulation, visualization, counterexamples, and candidate generation: operations that can expand the human's reasoning space while leaving evaluation human-owned. The execution shell contains boilerplate, formatting, syntax translation, UI alignment, repetitive transformations, and other work whose value is primarily operational. This layer is the strongest default target for automation.

Table 1. Example activities across the boundary.

Activity	Why it matters (or not)	Default treatment	When the boundary moves
Design a graph ontology	Creates the domain representation: entities, relations, granularity.	Retain / collaborate	May remain critical even for experts.
Interpret a surprising cluster	Builds causal intuition; tests assumptions against evidence.	Retain / collaborate	Automation can surface alternatives, not own the judgment.
Generate candidate schema variants	Expands the option space without requiring manual enumeration.	Collaborate	Automate generation; preserve comparison and justification.
Write a routine JOIN or endpoint	Often instantiates an already-understood pattern.	Collaborate $\rightarrow$ automate	High E for novices; low E after fluency.
Convert formats / rename fields	Primarily clerical transformation.	Automate	Retain only when data anomalies are themselves informative.
Adjust spacing / move a UI button	Execution detail unless interface design is the learning target.	Automate	Higher E for a novice specifically learning visual design.

4 The Boundary Must Move With Expertise

Recent programming studies illustrate why a fixed automation policy is unlikely to work. In a controlled study, access to an AI code generator improved novice code-authoring performance without a significant retention penalty [8]. Other work found that prior knowledge shaped who benefited: more prepared learners could use suggestions as accelerators, while struggling learners sometimes accumulated additional metacognitive difficulties and an illusion of competence [9]. CodeAid therefore deliberately avoided simply revealing complete code, and later work on cognitive-engagement techniques found value in requiring learners to reason through steps before code was revealed [10,11]. The evidence is not a blanket indictment of AI coding; it is an argument for state-dependent allocation.

Interactive concept: the boundary moves with expertise

The same activity can be knowledge-forming for a novice and routine execution for an expert.

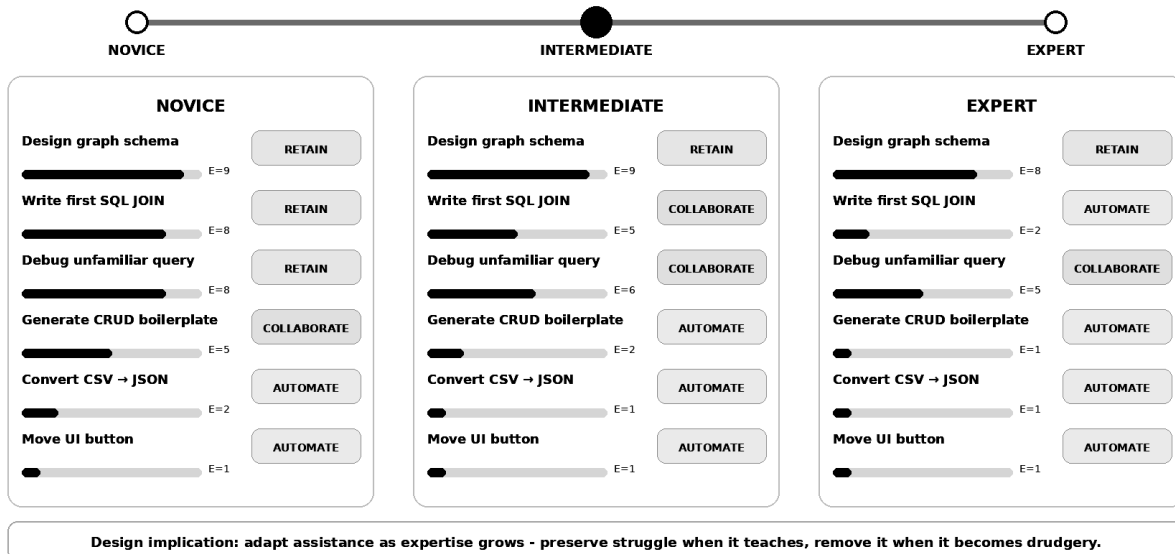


Figure 2. Interactive-style expertise slider. Illustrative E values show how the same activity can migrate from retain to collaborate to automate while high-level representation remains human-owned.

5 A Decision Framework for a Single Activity

Epistemic criticality should not be the only design variable. Let $X(a)$ denote execution burden - the time, attention, or fatigue consumed by an activity - and $A(a)$ the reliability with which the AI can perform it under the relevant conditions. The most attractive automation targets combine low E, high X, and high A. High-E activities should usually be retained or made collaborative even when AI is highly capable, because the goal is not to manufacture friction indiscriminately but to preserve cognition with future value. High-consequence domains add an obvious safety override: low epistemic value does not justify unchecked automation when errors are difficult to detect or costly.

Table 2. *Heuristic for drawing the boundary; it is a design aid, not a numerical scoring rule.*

Configuration	Interpretation	Recommended interaction
E high; A high	AI can do it, but doing it builds or exercises knowledge the human needs.	Retain human reasoning; let AI critique, compare, or check.
E medium; X high; A high	Some reasoning matters, but much of the workload is mechanical.	Collaborate: AI prepares candidates or partial work; human evaluates.
E low; X high; A high	The activity consumes effort without meaningful knowledge formation.	Automate aggressively; preserve lightweight verification.
E low; A low	Little learning value, but automation is not dependable.	Retain temporarily or redesign the task; do not confuse desirability with readiness.
E changes with H	The activity teaches novices but becomes routine with expertise.	Adaptive boundary: fade manual requirements as competence stabilizes.

6 What This Changes About Human-AI Teaming

First, it changes the object of optimization. A Human-AI team should be judged by two outputs: what the coupled system produces now and what capability remains with the human later. Human-centered AI guidelines emphasize control, feedback, and appropriate interaction [6]; cognitive-forcing work further shows that interfaces can deliberately require analytical engagement to reduce overreliance, even when users prefer lower-friction designs [7]. The Cognitive Automation Boundary makes this tension explicit: not all friction is waste, and not all convenience is benign.

Second, it changes what adaptive systems should adapt. Personalization should not only choose the best answer or the fastest workflow. It should decide how much of the reasoning process to expose, delay, require, or automate for this person at this moment. A novice might be asked to propose the relationship before seeing an AI-generated schema; an expert might receive the schema immediately and spend attention only on anomalies. The target is not permanent manual effort but a moving boundary that preserves productive cognition until it has done its developmental work.

Third, it changes organizational accounting. Productivity can rise while internal expertise quietly becomes thinner. Organizations adopting AI for engineering, analysis, research, or design should audit which recurring activities are building the mental models employees need to supervise AI, diagnose failures, transfer methods, and invent new approaches. A workflow that saves hours but removes every opportunity to form those models can create epistemic debt: capability resides increasingly in the coupled system while the human side becomes less able to interrogate it.

7 Research Agenda

The construct is deliberately testable. Researchers can estimate E indirectly through delayed unaided transfer, explanation quality, error detection, calibration, and the ability to reconstruct or adapt a solution after assistance is removed. Comparative studies can hold final task performance constant while varying which steps are automated. Longitudinal work can test whether boundaries should shift as expertise grows and whether users can productively control that shift themselves. Designers can prototype explicit boundary controls - sliders, staged reveal, candidate-before-answer interfaces, or "predict then inspect" interactions - and test when these mechanisms produce learning versus merely added workload.

8 Conclusion

The central question for Human-AI teaming is not simply which agent can perform each task better. It is which parts of the work must remain cognitively owned by the human for the human to remain capable of understanding, supervising, and extending the work. Epistemic criticality provides a language for making that distinction, and the Cognitive Automation Boundary turns it into a design decision. The boundary should be dynamic, expertise-sensitive, and explicit. Automate the labor that merely instantiates an idea; be cautious about automating the thinking through which the idea becomes understood. The question is not only what AI can think for us, but what humans must continue thinking about in order to remain knowledgeable.

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